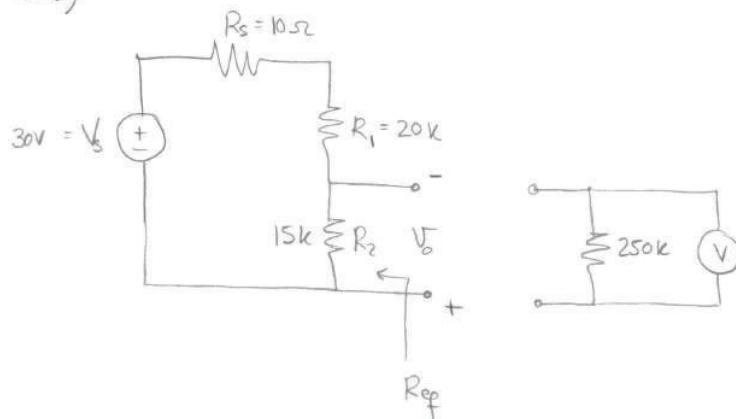


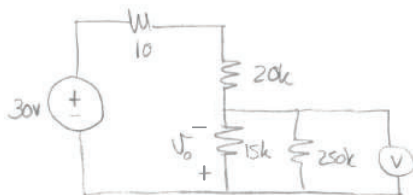
Q12)



i)
$$V_o = -30 \times \frac{15000}{10 + 15000 + 20000} = -12.85V$$

 voltage divider

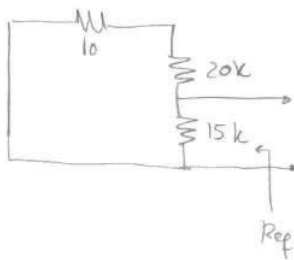
ii)



ii)
$$V_o = -30 \times \frac{15k \parallel 250k}{10 + 20k + 15k \parallel 250k} = -30 \times \frac{14150.94}{28581.4} = -12.43V$$

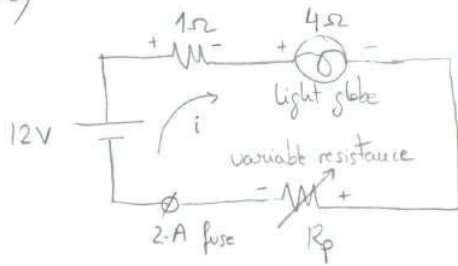
 voltage divider

iii)



iii)
$$R_{eq} = 15000 \parallel (20000 + 10) = 8573.26\Omega$$

Q1b)



i) $P_{\text{globe}} = 9\text{W} \Rightarrow i^2 \times 4 = 9 \Rightarrow i = \sqrt{9/4} = 1.5\text{A}$

KVL: $-12 + 1i + 4i + R_p i = 0$, where $i = 1.5\text{A}$

$$R_p = \frac{12 - 5 \times 1.5}{1.5} = \frac{4.5}{1.5} = 3\Omega$$

ii) Fuse will burn out for $i > 2\text{A}$

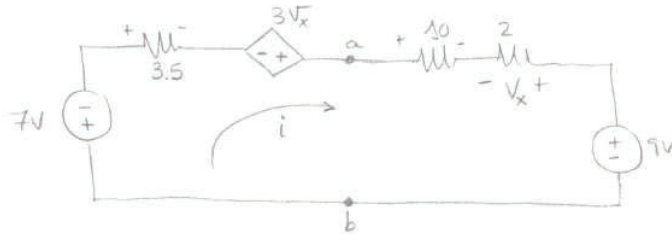
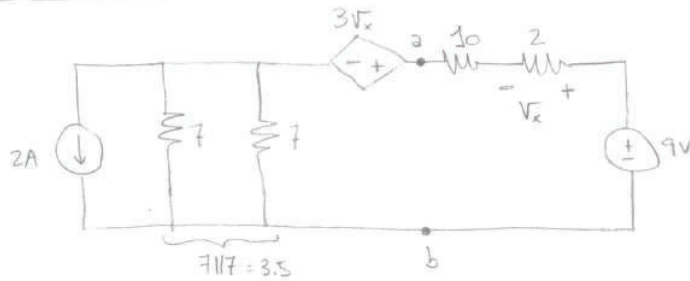
When $i = 2\text{A}$

KVL: $-12 + 1 \times 2 + 4 \times 2 + 2R_p = 0$

$$R_p = \frac{12 - 5 \times 2}{2} = \frac{2}{2} = 1\Omega \Rightarrow |R_p < 1\Omega| \text{ will burn out the fuse}$$

QUESTION 2a

i)



$$\text{KVL: } 7 + 3.5i - 3V_x + 10i - V_x + 9 = 0, \text{ where } V_x = -2i$$

$$7 + 3.5i + 6i + 10i + 2i + 9 = 0$$

$$16 + 21.5i = 0 \Rightarrow i = 16/21.5 = -0.7444 \text{ A}$$

$$\text{KVL on open mesh (left side): } 7 + 3.5(-0.7444) + 6(-0.7444) + V_{ab} = 0$$

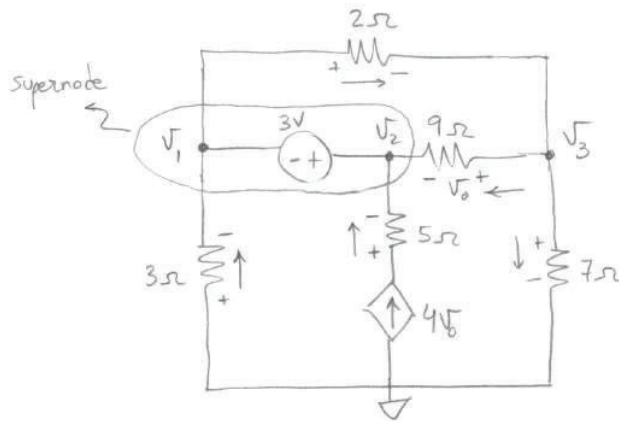
$$\boxed{V_{ab} = -7 + 2.604 + 4.464 = 0.068 \text{ V}}$$

$$\text{ii) } \boxed{P_{ds} = -V_{ds} \times i_{ds} = -4.464 \times (-0.7444) = +3.32 \text{ W}}$$

$$V_{ds} = 3V_x = 3(-2 \times (-0.7444)) = 4.464 \text{ V}$$

$$i_{ds} = i = -0.7444 \text{ A}$$

QUESTION 2b



* KCL @ supernode :

$$\frac{0 - V_1}{3} + 4V_0 + \frac{V_3 - V_2}{9} - \frac{V_1 - V_3}{2} = 0, \text{ where } V_0 = V_3 - V_2$$

* KCL @ node 3 :

$$\frac{V_1 - V_3}{2} - \frac{V_3 - V_2}{9} - \frac{V_3 - 0}{7} = 0$$

* Constraint equation :

$$V_2 - V_1 = 3$$

Simplify equations

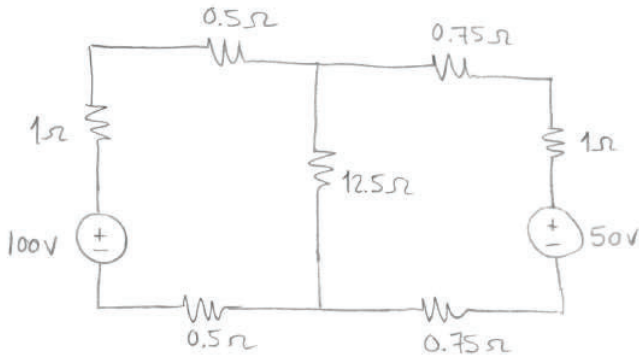
$$\begin{aligned} \textcircled{1} \quad & -\frac{V_1}{3} + 4(V_3 - V_2) + \frac{V_3 - V_2}{9} - \frac{V_1 - V_3}{2} = 0 \\ & -\frac{V_1}{3} + 4V_3 - 4V_2 + \frac{V_3}{9} - \frac{V_2}{9} - \frac{V_1}{2} + \frac{V_3}{2} = 0 \\ & \left(-\frac{1}{3} - \frac{1}{2}\right)V_1 + \left(-4 - \frac{1}{9}\right)V_2 + \left(4 + \frac{1}{9} + \frac{1}{2}\right)V_3 = 0 \\ & -0.8333V_1 - 4.111V_2 + 4.611V_3 = 0 \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad & \frac{V_1}{2} - \frac{V_3}{2} - \frac{V_3}{9} + \frac{V_2}{9} - \frac{V_3}{7} = 0 \\ & 0.5V_1 + 0.111V_2 - \left(0.5 + \frac{1}{9} + \frac{1}{7}\right)V_3 = 0 \\ & 0.5V_1 + 0.111V_2 - 0.754V_3 = 0 \end{aligned}$$

$$\textcircled{3} \quad V_2 - V_1 = 3$$

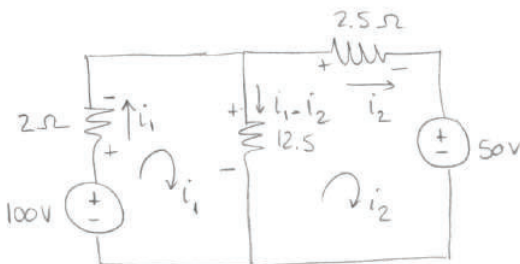
$$\begin{cases} -0.8333V_1 - 4.111V_2 + 4.611V_3 = 0 \\ 0.5V_1 + 0.111V_2 - 0.754V_3 = 0 \\ -V_1 + V_2 = 3 \end{cases}$$

QUESTION 3



i)
$$R_{eq} = \frac{250}{20} = 12.5 \Omega$$

ii) Resistors in series can be simplified (but even without simplification it is quite straight forward)



KVL @ mesh 1: $-100 + 2i_1 + 12.5(i_1 - i_2) = 0$

KVL @ mesh 2: $-12.5(i_1 - i_2) + 2.5i_2 + 50 = 0$

Simplify and solve for i_1 and i_2 :

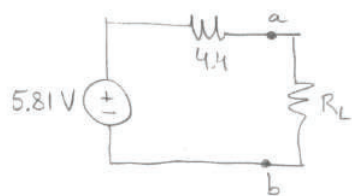
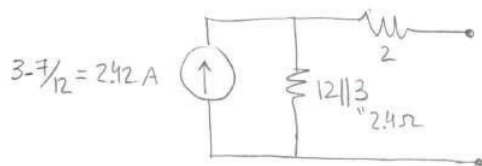
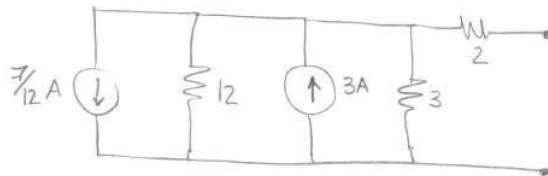
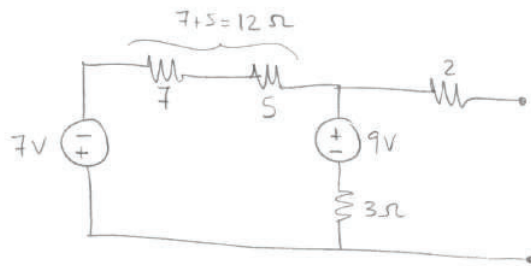
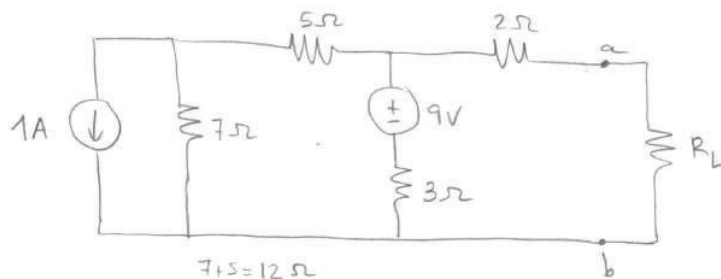
$$\begin{cases} 14.5i_1 - 12.5i_2 = 100 & (\times 15) \\ -12.5i_1 + 15i_2 = -50 & (\times 12.5) \end{cases} \Rightarrow \begin{cases} 217.5i_1 - 187.5i_2 = 1500 \\ -156.25i_1 + 187.5i_2 = -625 \end{cases}$$

$$61.25i_1 = 875 \Rightarrow i_1 = \frac{875}{61.25} = 14.29 \text{ A}$$

$$-12.5 \times 14.29 + 15i_2 = -50 \Rightarrow i_2 = \frac{-50 + 178.63}{15} = 8.575 \text{ A}$$

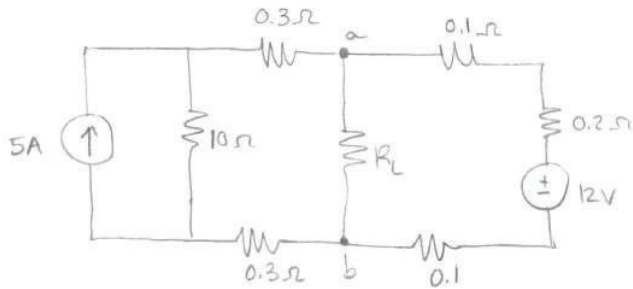
$$P_{\text{light}} = i_{\text{light}}^2 \times R = (i_1 - i_2)^2 \times 12.5 = (14.29 - 8.575)^2 \times 12.5 = 408.26 \text{ W}$$

QUESTION 42



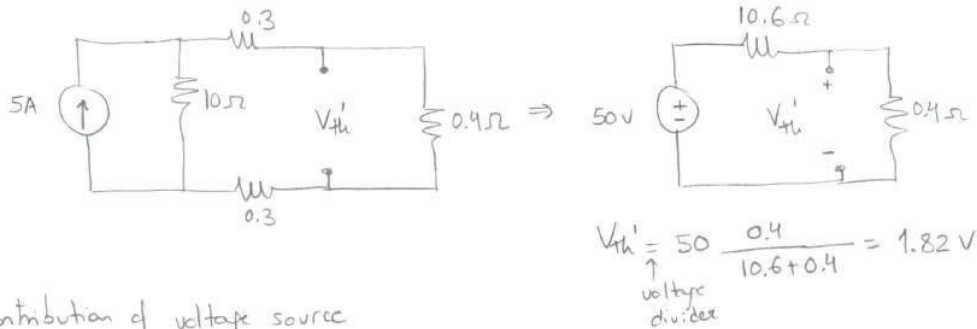
$$\Rightarrow \begin{aligned} V_{th} &= 5.81V \\ R_{th} &= 4.4\Omega \end{aligned}$$

QUESTION 4b



i) Use superposition to find V_{th} :

* Contribution of current source:

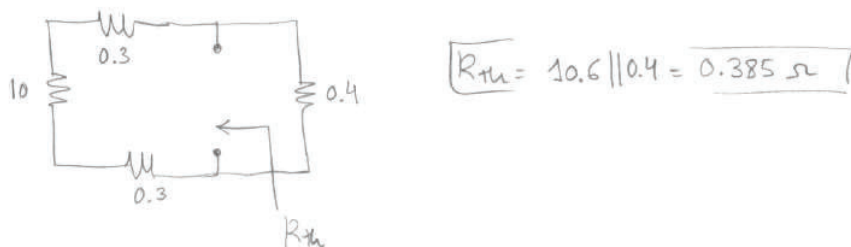


* Contribution of voltage source



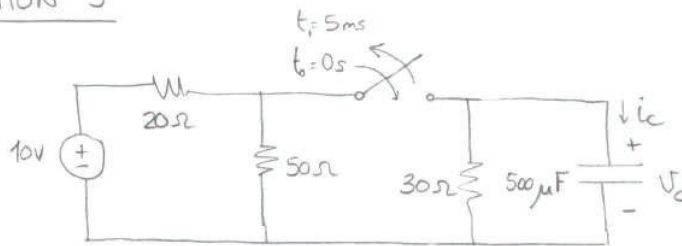
* Add contributions: $V_{th} = V_{th}' + V_{th}'' = 1.82 + 11.56 = 13.38 \text{ V}$

ii) Calculate R_{th} (method 1):



iii) $P_{max} = \frac{V_{th}^2}{4 R_{th}} = \frac{13.38^2}{4 \times 0.385} = 116.25 \text{ W}$

QUESTION 5



i) V_c for $0 \leq t \leq 5 \text{ ms}$



Step response

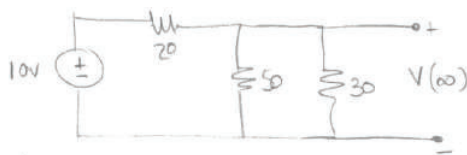
$$V_c = V(\infty) + [V(0) - V(\infty)] e^{-t/\tau}$$

where $\tau = R_{Th} C$

* Calculate $V(0)$ for $t \leq 0$

$V(0) = 0 \text{ V}$ (switch open for a long time $\Rightarrow$ capacitor discharged)

* Calculate $V(\infty)$ for $0 \leq t \leq 5$



voltage divider

$$V(\infty) = 10 \frac{50 \parallel 30}{20 + 50 \parallel 30} = 4.839 \text{ V}$$

* Calculate τ :

Calculate R_{Th} :



$$R_{Th} = 20 \parallel 50 \parallel 30 = 9.678 \Omega$$

$$\tau = R_{Th} C = 9.678 \times 500 \mu\text{F} = 0.004839 \text{ s} = 4.839 \text{ ms}$$

* Substitute:

$$V_c = 4.839 + [0 - 4.839] e^{-t/0.004839} = 4.839 (1 - e^{-206.65 t}) \text{ V}$$

ii) V_c for $t \geq 5 \text{ ms}$



Natural response

$$V_c = V(0.005) e^{-(t-0.005)/\tau} \text{ where } \tau = RC$$

* Calculate $V(0.005)$ from previous result:

$$V(0.005) = 4.839 (1 - e^{-206.65 \times 0.005}) = 3.117 \text{ V}$$

* Calculate τ :

$$\tau = RC = 30 \times 500 \mu\text{F} = 0.015 = 15 \text{ ms}$$

* Substitute:

$$V_c = 3.117 e^{-(t-0.005)/0.015} = 3.117 e^{-66.667 (t-0.005)} \text{ V}$$

$$\begin{aligned} \text{iii)} \quad i_c &= C \frac{dv_c}{dt} = 500 \mu\text{F} \times 3.117 \times (-66.667) e^{-66.667(t-0.005)} \\ &= -0.104 e^{-66.667(t-0.005)} \end{aligned}$$

$$\boxed{i_c(10\text{ms}) = i_c(0.01) = -0.104 e^{-66.667(0.01-0.005)} = -0.075 \text{ A}}$$